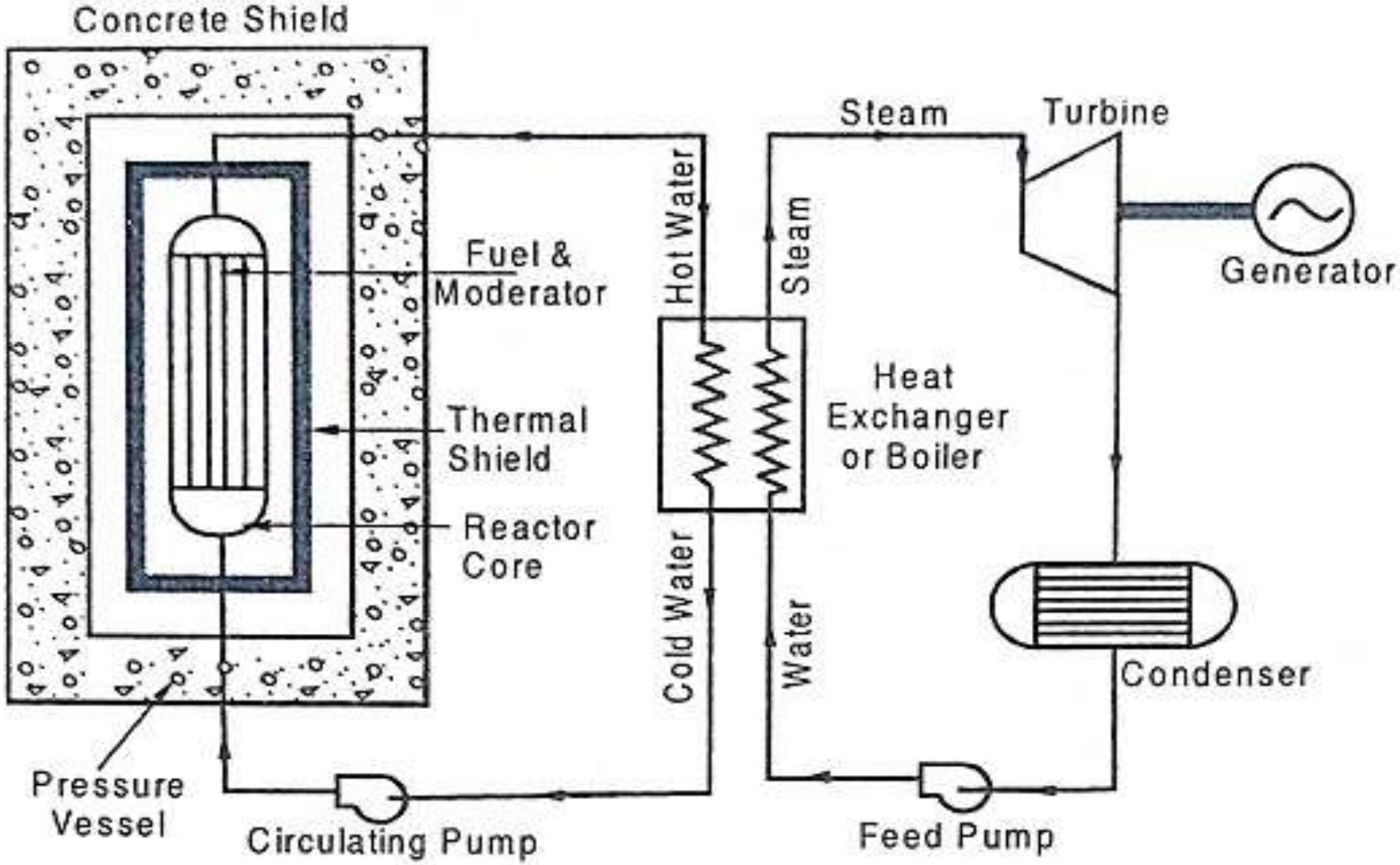
COMPONENTS OF NUCLEAR REACTOR


Concrete Shield
Fuel &
Moderator
Thermal
Shield
Reactor
Core
Hot Water
Steam
Steam
Turbine
Generator
Heat
Exchanger
or Boiler
Cold Water
Water
Condenser
Pressure
Vessel
Circulating Pump
Feed Pump

1. Fuel Rods

- Fuel rods contain the fissile material (usually uranium-235 or plutonium-239) where the nuclear fission reactions occur.
- Fuel rods are thin tubes that contain nuclear fuel, usually in the form of uranium-235 or plutonium-239 pellets, where the nuclear fission reactions take place. These rods are assembled into bundles, forming the reactor core's fuel assemblies. The fission of atoms within these rods releases large amounts of heat, which is used to generate electricity. Fuel rods are typically made of materials like zirconium alloy, which can withstand high temperatures and is resistant to corrosion, ensuring they perform effectively within the intense environment of a nuclear reactor.

2. **Moderator** - A moderator is a material in a nuclear reactor that slows down the fast neutrons produced during fission to increase the likelihood of further fission reactions. By slowing neutrons to thermal speeds, the moderator helps sustain the chain reaction within the reactor.

Types of moderator

- Light Water (H_2O) - Used in Pressurized Water Reactors (PWRs) and Boiling Water Reactors (BWRs) when Enriched uranium is used as fuel.
- Heavy Water (D_2O) – in above reactors when natural uranium used as fuel.
- Graphite - Used in gas-cooled reactors and Liquid Metal cooled Reactor (LMCR.)

3. Control Rods

Control rods are essential components in nuclear reactors used to control the rate of the fission chain reaction. They are made of materials that absorb neutrons, such as boron, cadmium, or hafnium. By absorbing neutrons, control rods reduce the number available to sustain the chain reaction. They can be inserted deeper into the reactor core to slow or stop the reaction or raised to increase it. This adjustability allows operators to maintain a stable and controlled reaction, ensuring the reactor operates safely and effectively.

Material: Control rods are typically made of materials with high neutron absorption capabilities, like boron, cadmium, hafnium, or silver-indium-cadmium alloys.

4. Pressure Vessel

The pressure vessel in a nuclear reactor is a robust, thick-walled steel container that houses the reactor core, coolant, and, in some designs, the moderator.
Primary function –

- ❖ Maintain high pressure
- ❖ Keeps the coolant in a liquid state at elevated temperatures which is crucial for efficient heat transfer from the reactor core.

5. Coolant

Coolant in a nuclear reactor is essential for absorbing and transferring the heat generated by fission in the reactor core, either to produce steam for power generation or to prevent overheating.

Coolants being used - Light Water (H_2O), Heavy Water (D_2O)
Carbon Dioxide (CO_2), Helium , Liquid Sodium ,

6. Reactor Core

The reactor core is the central part of a nuclear reactor where the nuclear fission reactions occur. It is the heart of the reactor and contains the fuel rods, moderator, and control rods, all working together to sustain a controlled nuclear chain reaction.

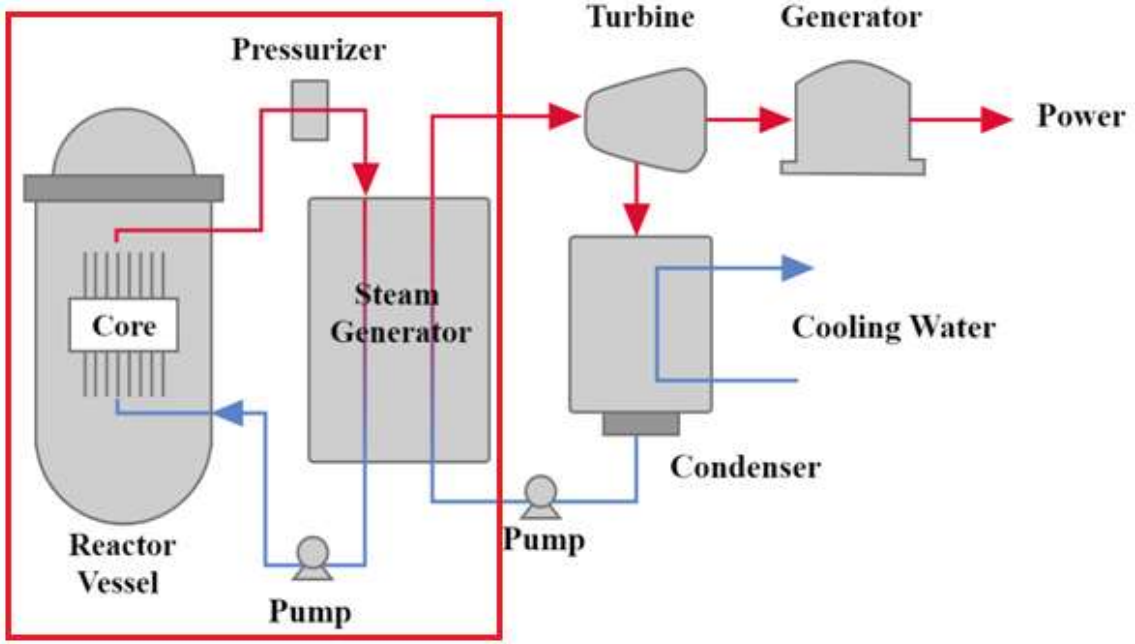
Key Features of the Reactor Core:

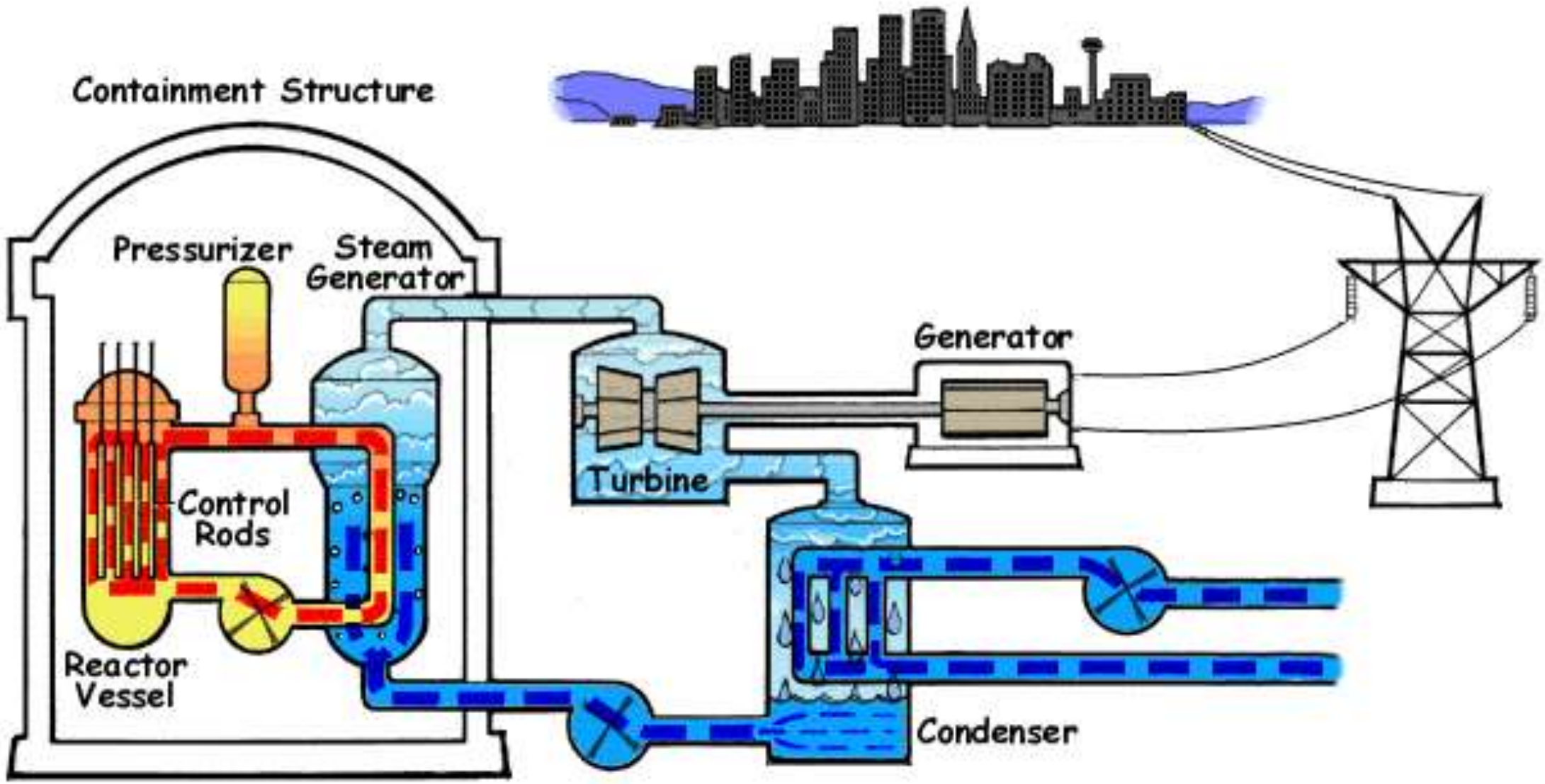
1. **Fuel Assemblies**: The core is composed of multiple fuel assemblies, which contain the fuel rods filled with fissile material, typically uranium or plutonium.
2. **Moderator**: The moderator is present within the core to slow down fast neutrons produced by fission, making them more likely to induce further fission reactions. Common moderators include water (light or heavy) and graphite.
3. **Control Rods**: Control rods are inserted into or withdrawn from the core to manage the fission reaction rate. They absorb neutrons and help maintain a stable chain reaction.
4. **Coolant Flow**: The core is surrounded by coolant, which absorbs the heat generated by fission. This coolant is circulated to transfer heat away from the core to produce steam for electricity generation.
5. **Containment**: The core is housed within a pressure vessel, designed to withstand high pressure and temperature, ensuring the containment of radioactive materials.

Pressurized Water Reactor (PWR)

- **Working**

  - The coolant water is pumped to the reactor core where the coolant gets heated by the nuclear fission and the hot coolant is feeds to the pressurizer.
  - The pressurized hot fluid is passes to the steam generator where the steam is generated by the energy transfer from hot fluid to the feed water.
  - The steam gets expanded in the steam turbine and produce the mechanical power output at the shaft.
  - The expanded steam is condensed in the condenser and pumped back to the steam generator.
  - The alternator is coupled to the turbine to generate the electricity.




Containment Structure
Pressurizer
Steam Generator
Control Rods
Reactor Vessel
Turbine
Generator
Condenser

Boiler Water Reactor (BWR)

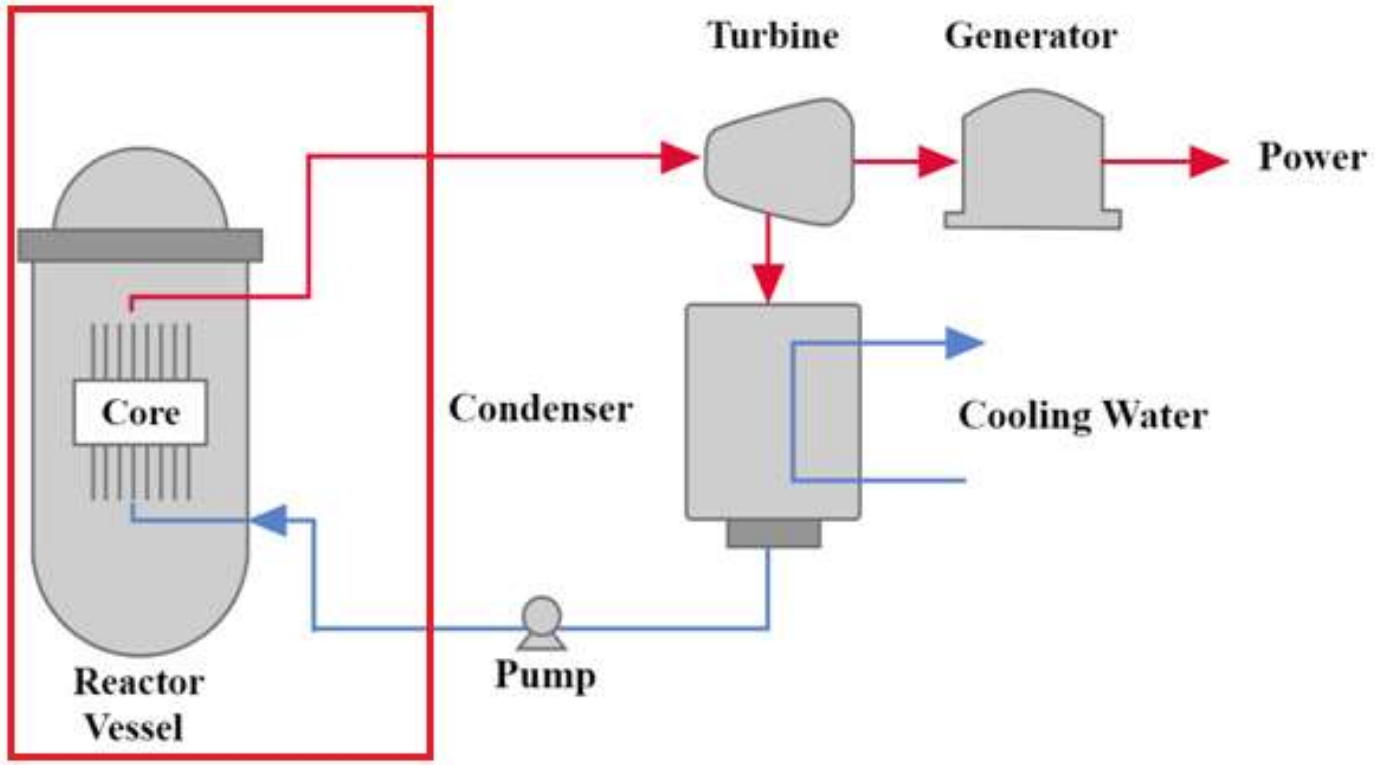
- **Working**

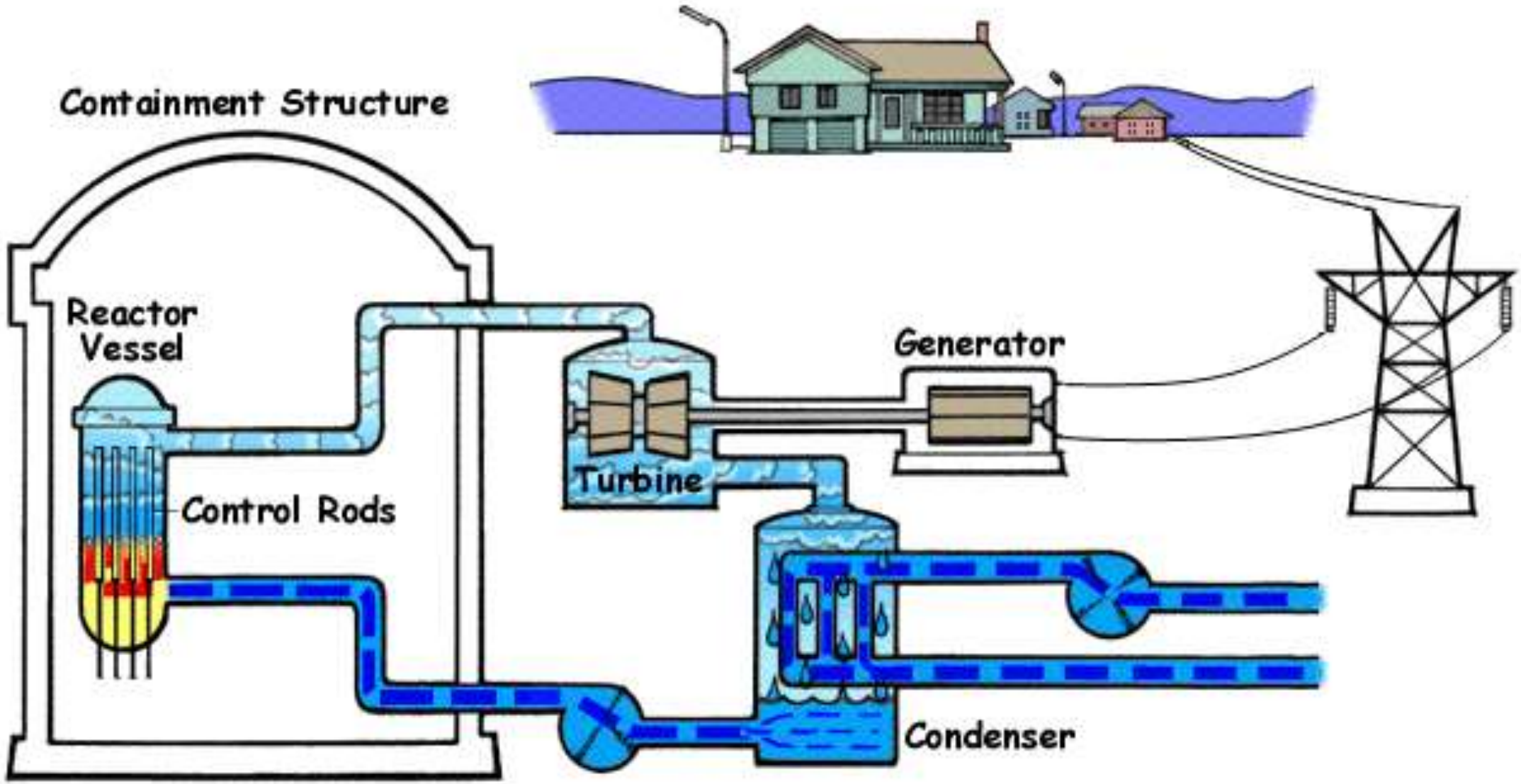
➢ The only difference between PWR and BWR is the steam is generated in the reactor instead of using steam generator.

➢ Water is pumped up through the reactor core and heated by fission. reactor heats water and produce steam directly inside the reactor vessel.

➢ The steam leaves the top of the reactor, passes through the steam turbine and gets expanded.

➢ The expanded steam is then condensed back to water and pumped to the reactor.




Containment Structure
Reactor Vessel
Control Rods
Turbine
Generator
Condenser